CORVUS SENTINEL · SHOTS FIRED

Pricing, Margin & Acquisition Model

INTERNAL / PARTNER COPY — contains OCWS margin data.

How we undercut ShotSpotter with owned hardware + recurring service + the Crow's Eye survey, outsource install and maintenance, and still profit — worked against the Escambia County program. The model and its levers, not a committed bid.

01 What we're undercutting

ShotSpotter is a per-square-mile annual subscription, always rented, never owned: **$65K-$90K per mi²/yr + ~$10K/mi² one-time** (secondary); **NYC $85,097/mi²/yr, 3-yr term** (verified). Because of the price, cities can only afford small hot-zones (a few to ~50 mi²) and own nothing at the end. The wedge: owned hardware, county-wide, for a fraction of the multi-year rent.

02 Build cost per node (materials + build labor)

Hand-built now (~$25/unit at ~20 units/person/day; validate with a pilot); contract-manufactured later ($15-40/unit consignment, NRE negligible at volume). Note: the Pi Zero has ~0% volume discount (retail is subsidized) — a real exception to plan for.

Node type	Materials	+ Build	Cost
Indoor leaf	$150	$25	$175
Indoor gateway	$230	$25	$255
Roof gateway	$270	$30	$300
Vehicle	$290	$30	$320
Drone	$55	$15	$70

03 Escambia program — three revenue lines

Node counts: 952 leaves, 100 gateways, 250 vehicles, 10 drones (~90 buildings).

Revenue line	Detail	Customer	Gross
Hardware (capex)	cost $272,800, sell 1.25x	~$341,000	~$68,000
Install (outsourced)	cost $206,500, +1.3x markup	~$268,000	~$62,000
Crow's Eye survey	~90 buildings, ~73% margin	~$99,000	~$72,000
One-time subtotal		~$708,000	~$202,000
Annual service (x5 yr)	C2 + monitoring + maint.	~$750,000	~$250,000
5-YEAR TOTAL		~$1,458,000	~$452,000

Blended 5-yr margin ~31%. Hardware near-cost to win; profit is survey + recurring. Service cost ~$100K/yr (RMM $3/device/mo x1,300 + hosting + support/truck rolls); priced $150K/yr with room to $200K+ given the ShotSpotter gulf. Battery cadence is the #1 maintenance lever.

04 5-year TCO vs ShotSpotter

	Coverage	5-year cost	Owned?
ShotSpotter (20-50 mi²)	outdoor gunshot, hot-zones	~$7M - $18M	No
Sentinel (county-wide)	interiors + county + mobile	~$1.5M	Yes

~5-12x cheaper over five years, more coverage, owned. Plus auditable accuracy and one command picture (drones + gunshots + mobile).

05 Acquisition timeline (close -> installed)

Stage	Hand-build (pilot)	CM (county scale)
Procure components	2-6 wk	4-12 wk
Assemble + QA	~2-4 wk	4-8 wk (8-12 first)
Survey + schedule	3-6 wk	4-8 wk
Install	2-4 wk	6-16+ wk
Net close -> installed	~2-4 months	~4-7 months

06 Supply lines & the four numbers to close

Shortage is over (2021-23); 2025-26 headwind is memory-price inflation, not availability. Verified: SIM7080G cellular ~$8.50/unit @ 1k, ~2-wk lead; Pi Zero ~$15 flat (no volume discount). Verify-live: Pi 4/CM4, SX1262 LoRa, MEMS mic, 18650/LiFePO4 cells. CM: NRE $600-1,750, box-build $15-40/unit, 4-8 wk (8-12 first build).

Close before it's a bid: (1) live volume component prices/lead-times; (2) real per-vehicle upfit labor; (3) CM per-unit quote; (4) ECSO fleet count.